Integrated Student Organization Activity Approval and Fund Release Platform

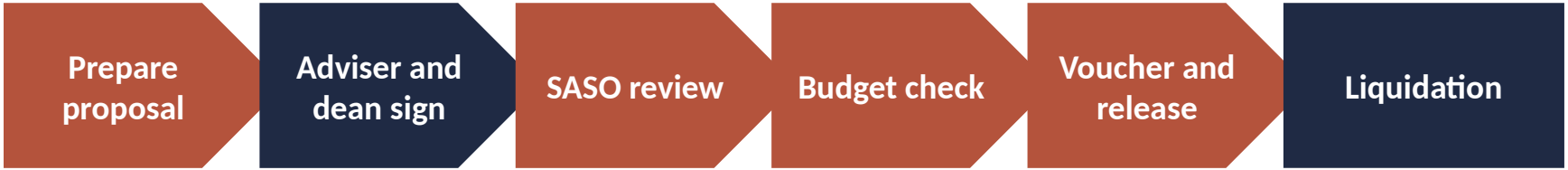
Prefinal Project Proposal · ITPC 313 – System Integration and Architecture

Project Lead / Architect: Jessa Angela Sabueto
Integration Developer: Alex Mendoza Jr.
Data and Security Lead: Joey Manlapas
QA and Documentation Lead: Faith Red
Team FAQsilica · Divine Word College of Legazpi · November 2026

The problem

Slide 2 · 0:45

Presenter: Faith Red



Highlighted steps: data is retyped, checked by hand, or checked too late

Delays

Status known only by asking

Errors

Details retyped at each step

Duplicate work

Budget checked after approval

Systems involved

Slide 3 · 0:45

Presenter: Alex Mendoza Jr.

System	Owner	Data it holds	Interface
SORS registry	SASO	Orgs, officers, accreditation	REST
AAWS approval	SASO, deans	Proposals, decisions	REST; sends ActivityApproved
FBS finance	Finance Office	Balances, vouchers	REST; sends FundReleased
SIS (mocked)	Registrar	Enrollment status	REST lookup
Notification (mocked)	IT Services	Message templates	Reads events

Objectives and scope

Slide 4 · 1:00

Presenter: Joey Manlapas

General objective: one automated, secure, traceable flow from approval to fund release

0	fields typed twice
100%	over-budget requests rejected
95%	disbursed within 3 s at 50 users
0	lost or duplicate events
100%	status changes logged

In scope
REST, events, gateway, canonical model, tests

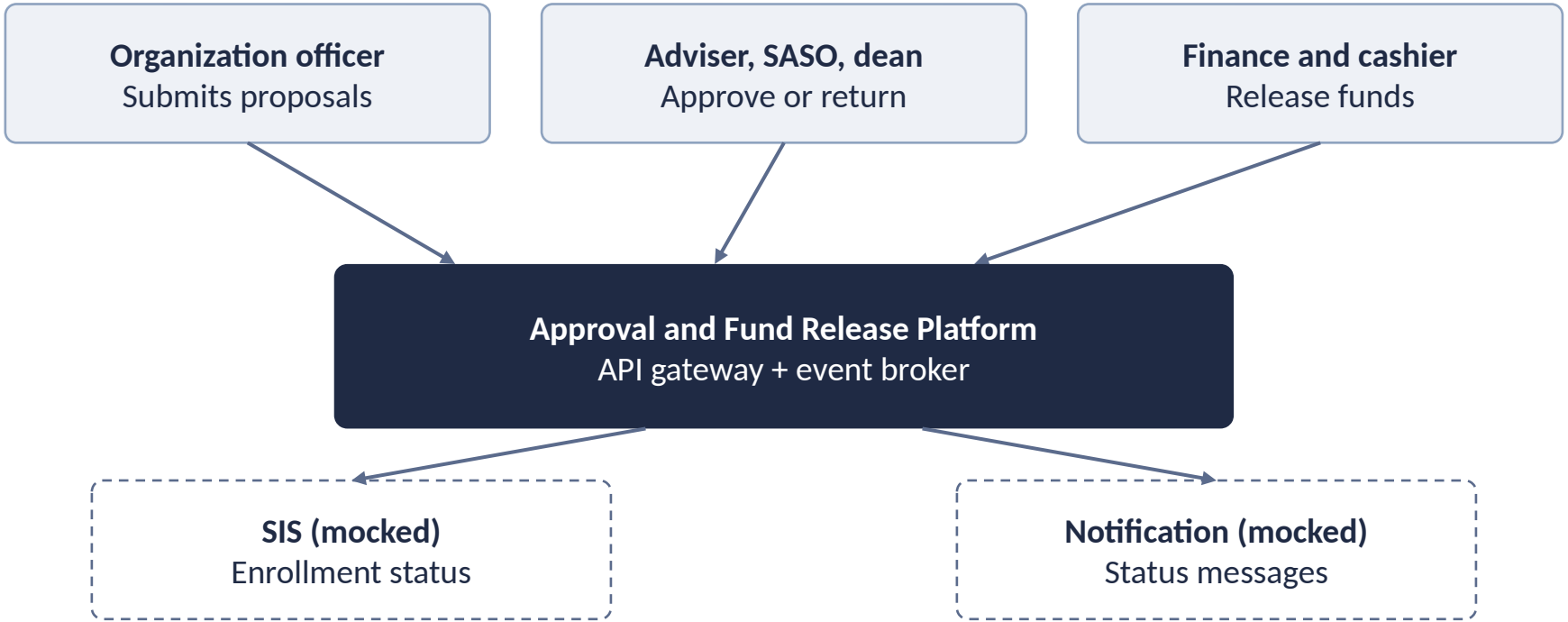
Out of scope
Real payments, real data, mobile app

Mocked
SIS, notifications, cashier, sign-on

C4 Context diagram

Slide 5 · 1:15

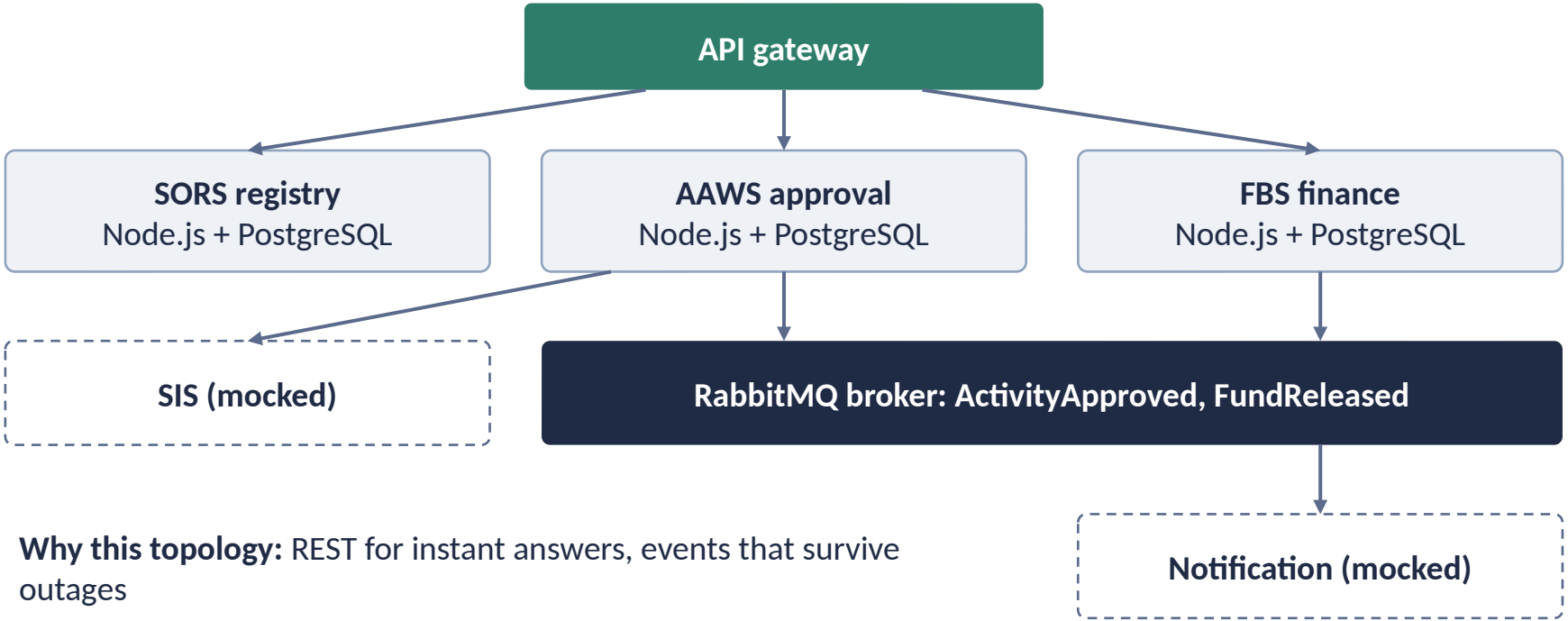
Presenter: Jessa Angela Sabueto



C4 Container diagram and topology

Slide 6 · 1:15

Presenter: Jessa Angela Sabueto



Why this topology: REST for instant answers, events that survive outages

Integration flows

Slide 7 · 1:30

Presenter: Alex Mendoza Jr.



Canonical data model

One JSON Schema per event; money in centavos; UTC timestamps

Dark boxes: REST calls. Green boxes: events through the broker

Security, privacy, and reliability

Slide 8 · 1:00

Presenter: Joey Manlapas

Threat	Control
Other org's record	Org ID in token
Forged token	Signed 15-min JWT
Overload	Gateway rate limit
Bad event data	JSON Schema check
Personal data	Student ID only (RA 10173)

If this fails	What happens
Broker down	Saved, sent later
FBS down	Queued, replayed
Same message twice	Ignored by eventId

Technology stack and testing plan

Slide 9 · 1:00

Presenter: Faith Red

Layer	Tool
Services	Node.js, Express
Broker	RabbitMQ
Database	PostgreSQL
Containers	Docker Compose
Tests	Jest, k6

Test	Pass when
Flow test	Submit to release works
Load test	95% under 3 s, 50 users
Failure test	0 events lost
Replay test	1 request per 1,000 repeats

Work plan and top risks

Slide 10 · 1:00

Presenter: Faith Red

Task	P1	P2	P3
Scenario, repository (Jessa)	■		
Parts 1-7 draft (Faith)	■		
Diagrams, ADRs (Jessa)		■	
API, events, schemas (Alex)		■	
Security, privacy (Joey)		■	
Full proposal (Faith)		■	
Slides, rehearsal (all)			■

P = Prefinal week

Top 3 risks

Set-up takes long: Compose skeleton first

Scope creep: scope table is the baseline

Late security flaws: security tests before code

Questions

Thank you · 10:00 of questions from the instructor and classmates